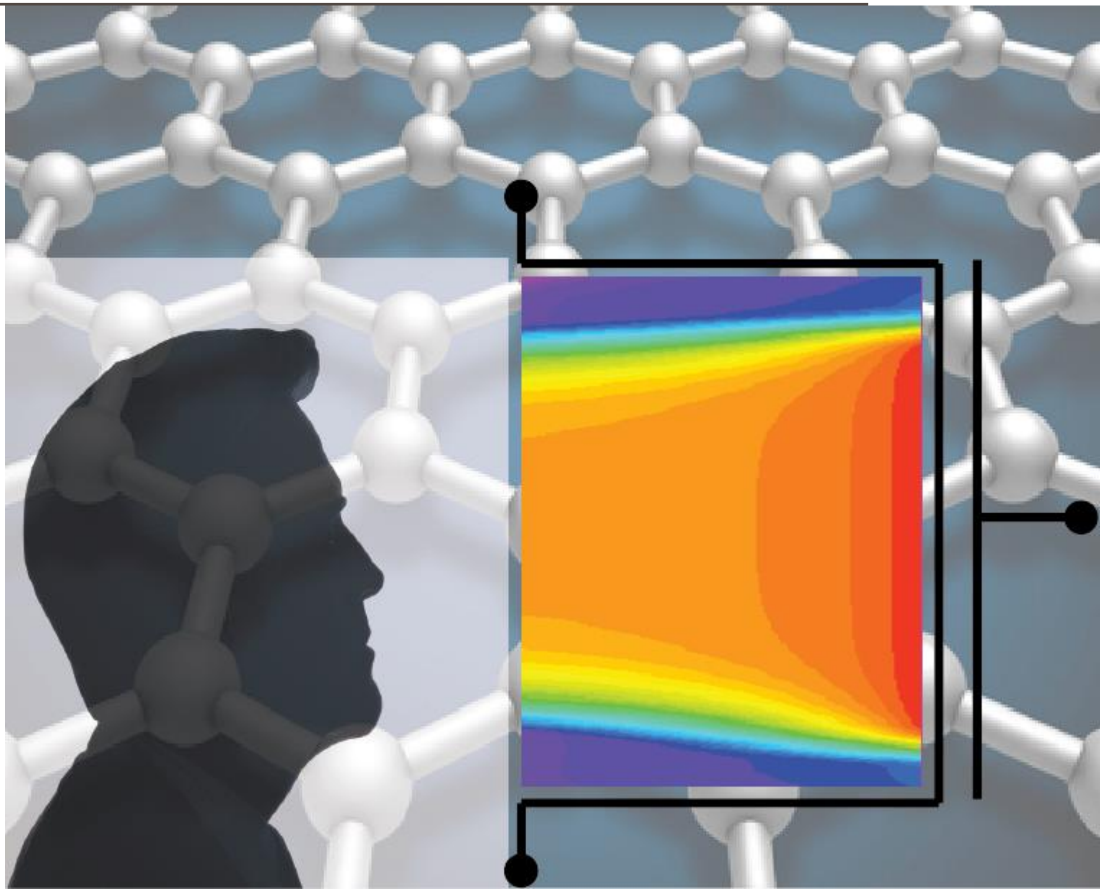
ADVANCED SOLID-STATE DEVICES

Week-7, Lecture-2



Sneh Saurabh
20th February, 2025

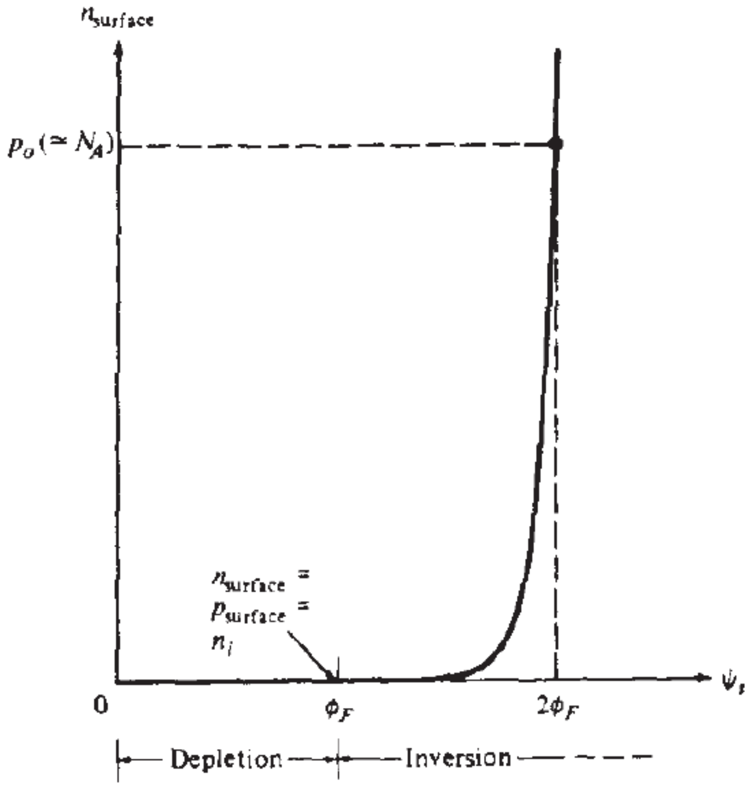
MOSFETs

MOS Capacitor: Surface Potential vs. Electron Concentration at Surface (2)

$$n_{surface} = n_i\ exp\left(-\frac{\phi_{Fp} - \psi_s}{kT/q}\right) = N_A exp\left(\frac{\psi_s - 2\phi_{Fp}}{kT/q}\right)$$

- At $\psi_s = \phi_{Fp}$, $n_{surface} = n_i$ (surface becomes intrinsic)
- This point is the change from depletion to inversion
- When increases more than this point, $n_{surface}$ increases sharply

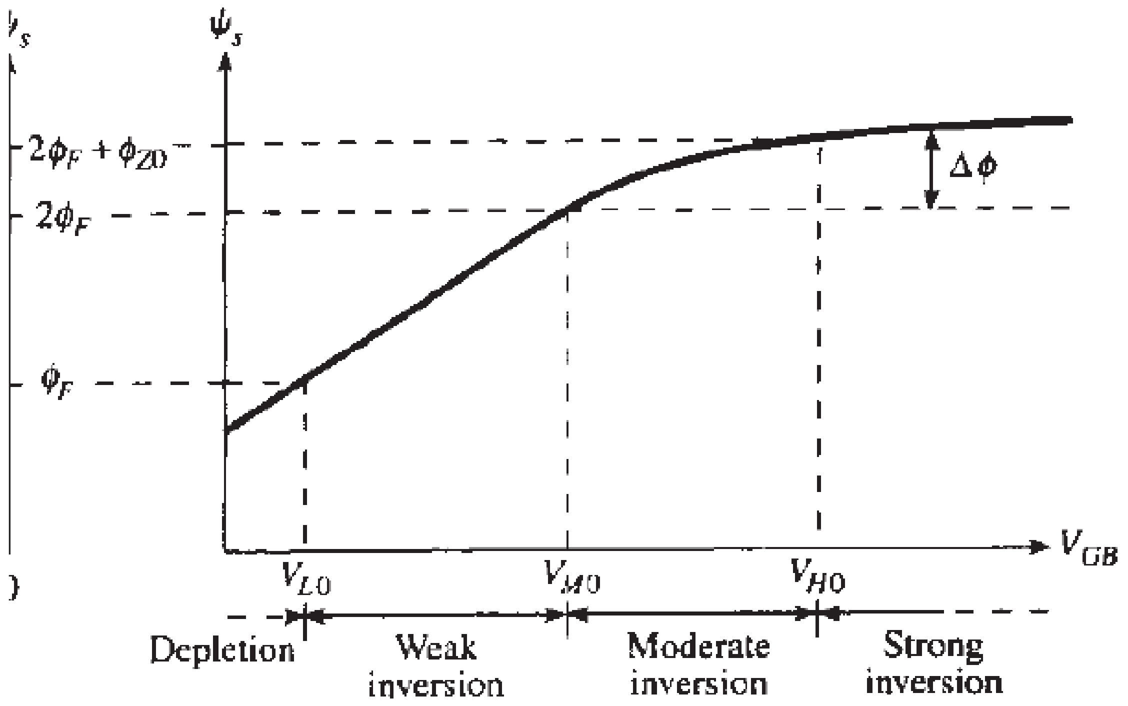
- At $\psi_s = 2\phi_{Fp}$, $n_{surface} = N_A$ (surface has same carrier concentration as bulk, but with opposite type)
- The surface considered to be strongly inverted



Source: Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.

MOS Capacitor: Surface Potential vs. Gate Voltage

- The surface potential increases with gate voltage
- At much higher gate voltage, increase in gate voltage does not increase surface potential (only charge in the inversion region increases)
- Traditionally, $2\phi_F$ is taken as a ψ_s at which the increase in surface potential becomes very slow
- This point is traditionally taken as ***threshold voltage point***



Source: Tsividis, Yannis. *Operation and Modeling of the MOS Transistor.* McGraw-Hill, Inc., 1987.

MOS Capacitor: Depletion Layer Width

Problem: Assume that the surface potential is ψ_s. Compute the width of the depletion region using one-sided n⁺p junction approximation.

Depletion region width: $W = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\left[\frac{(N_a+N_d)}{N_a N_d}\right]}$

For n⁺p junction $W = \sqrt{\frac{2\varepsilon_s V_{bi}}{qN_a}}$

Depletion layer width in MOS capacitor: $x_d = \sqrt{\frac{2\varepsilon_s \psi_s}{qN_a}}$

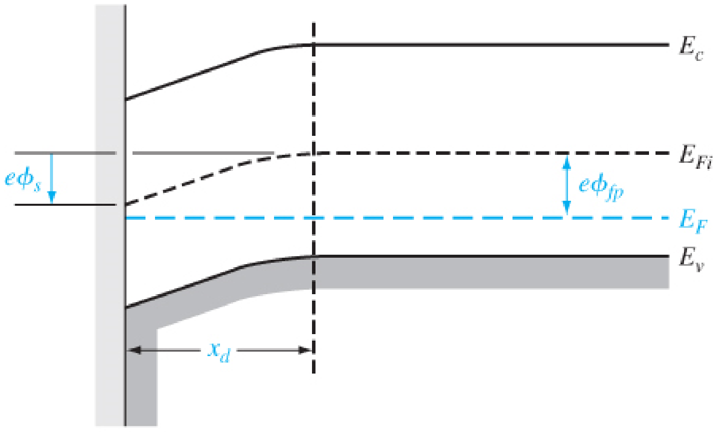


Figure 10.8 | The energy-band diagram in the p-type semiconductor, indicating surface potential.

MOS Capacitor: Maximum Depletion Layer Width

The maximum depletion layer width is reached when $\psi_s = 2\phi_F$.

Depletion layer width in MOS capacitor:

$$x_d = \sqrt{\frac{2\varepsilon_s \psi_s}{qN_a}}$$

So, the maximum depletion layer width in MOS capacitor:

$$x_{d,max} = \sqrt{\frac{4\varepsilon_s \phi_F}{qN_a}}$$

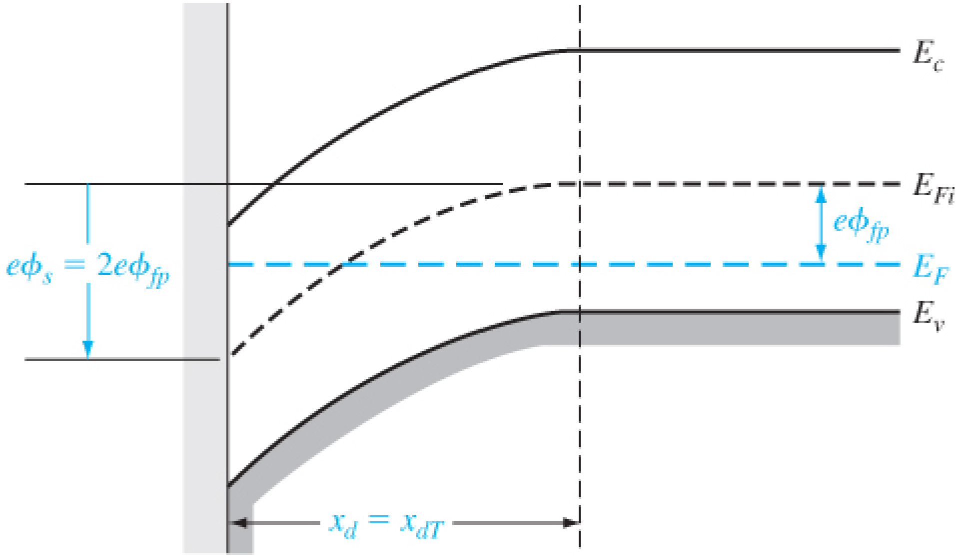
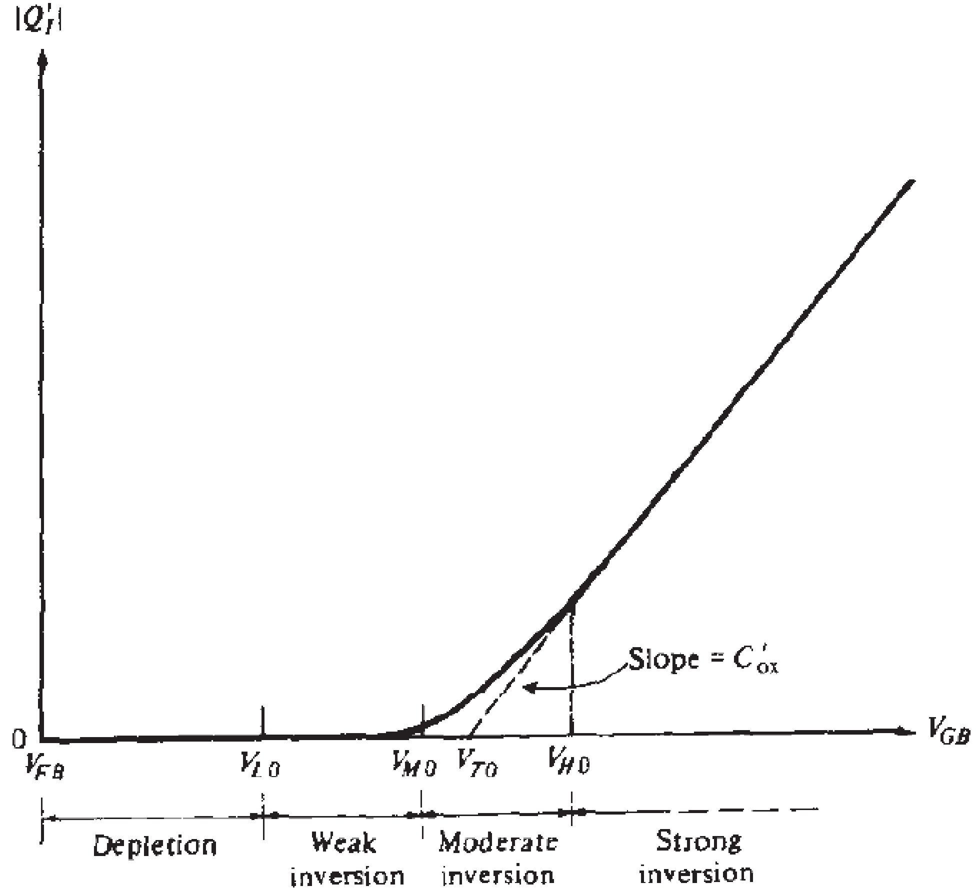


Figure 10.9 | The energy-band diagram in the p-type semiconductor at the threshold inversion point.

MOS Capacitor: Inversion Layer charge (1)

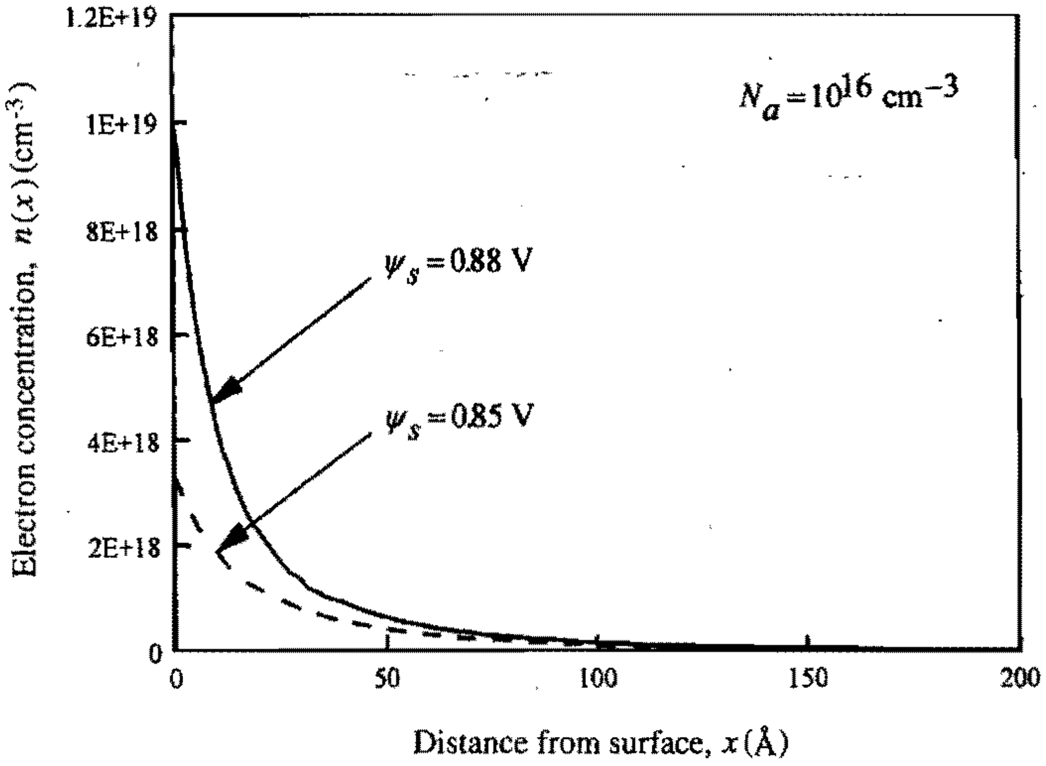
- When the MOS capacitor reaches strong inversion, the depletion layer width remains constant
- The extra voltage applied leads to increase in inversion layer charge



Source: Tsividis, Yannis. *Operation and Modeling of the MOS Transistor.* McGraw-Hill, Inc., 1987.

MOS Capacitor: Inversion Layer charge (2)

- Charge in the inversion layer is mostly situated very close to the surface
- Electrons confined to around 50 Å (can be considered as 2D electron gas and quantum confinement effects observed)
- Increase in surface potential ψ_s leads to more confinement
- MOS with inversion layer behaves as parallel plate capacitor

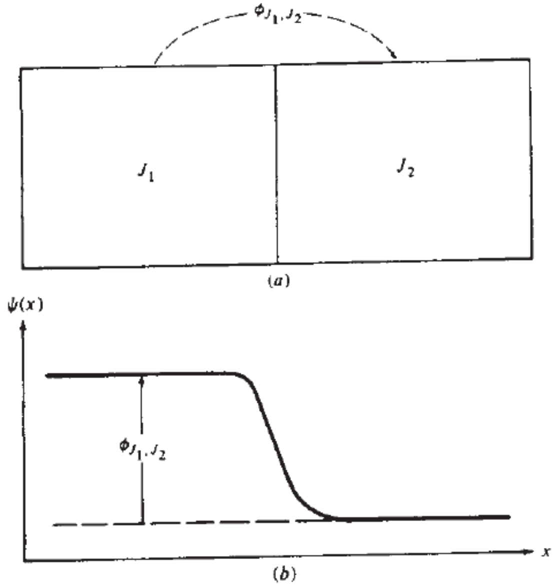


Source: Taur, Yuan, and Tak H. Ning. *Fundamentals of modern VLSI devices*. Cambridge university press, 2021.

Concept of Contact Potential

Contact Potential: Definition

Consider two materials J_1 and J_2. Assume that they are made in contact. What would you expect in equilibrium?



- Charge carriers are at different energy in two materials
- They will move until electric field establishes and opposes the motion

An electrostatic potential change is encountered in going from one material to other

Total potential drop in going from J_1 and J_2 is called ***contact potential*** (denoted as $\phi_{J_1 J_2}$)

Contact potential is given by:

$$\phi_{J_1 J_2} = \phi_{J_1} - \phi_{J_2}$$

where ϕ_{J_1} is the contact potential with respect to some reference and ϕ_{J_2} is contact potential with respect to same reference

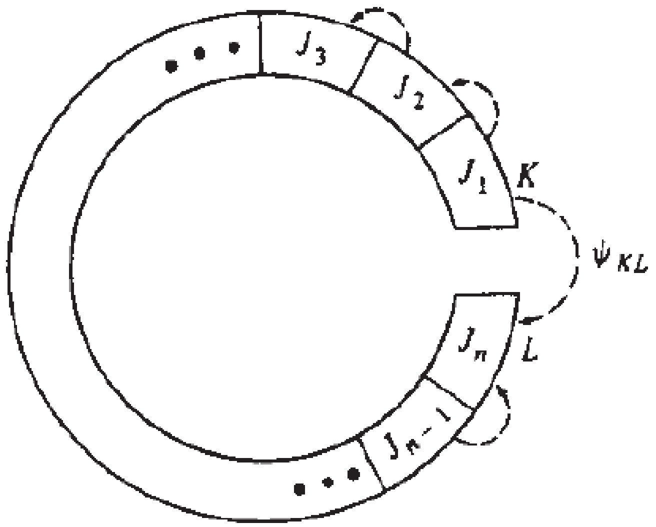
- Any convenient reference can be taken
- For example, vacuum level or intrinsic silicon level

Contact Potential: Materials in Series

Problem: Consider n materials $J_1, J_2., \ldots, J_n$ in contact as shown (all materials are at the same temperature). Assume that the contact potential of each material is ϕ_{J_1} , ϕ_{J_2} , $\ldots \phi_{J_n}$ with respect to some common reference. Compute the potential ψ_{KL}.

Potential is given by:

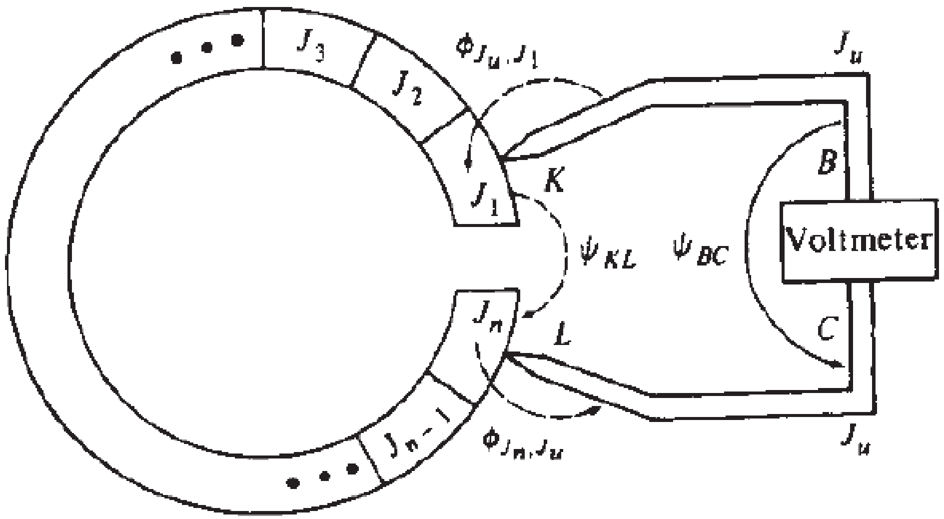
$$\psi_{KL} = \phi_{J_1 J_2} + \phi_{J_2 J_3} + \cdots + \phi_{J_{n-1} J_n}$$
$$= (\phi_{J_1} - \phi_{J_2}) + (\phi_{J_2} - \phi_{J_3}) + \cdots + (\phi_{J_{n-1}} - \phi_{J_n})$$
$$= \phi_{J_1} - \phi_n$$



- No matter how many materials are there in the loop, the electrostatic potential difference between its two ends depends *only* on the first and the last material.

Contact Potential: Can you measure it?

Problem: Consider a series of material in contact as described in the last problem is connected with an ideal voltmeter. Assume that the voltmeter leads are made of some material with contact potential ϕ_{J_u}. What will be the measured potential ψ_{BC}?



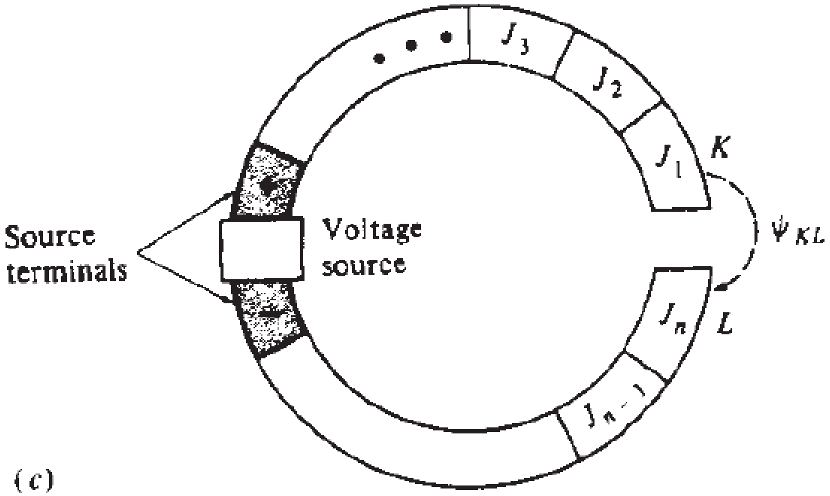
Potential is given by:

$$\psi_{BC} = \phi_{J_uJ_1} + \phi_{J_1J_2} + \phi_{J_2J_3} + \cdots + \phi_{J_{n-1}J_n} + \phi_{J_nJ_u}$$
$$= (\phi_{J_u} - \phi_{J_1}) + (\phi_{J_1} - \phi_{J_2}) + (\phi_{J_2} - \phi_{J_3}) + \cdots$$
$$+ (\phi_{J_{n-1}} - \phi_{J_n}) + (\phi_{J_n} - \phi_{J_u})$$
$$\psi_{BC} = 0$$

- An ideal voltmeter will not see any voltage difference and we cannot measure the contact potential

Contact Potential: Invisible?

Consider that a voltage source V_{source} is inserted in the loop. What would an ideal voltmeter read the voltage between J_1 and J_n



(c)

Potential is given by:

$$\psi_{BC} = \phi_{J_uJ_1} + \phi_{J_1J_2} + \phi_{J_2J_3} + \cdots V_{source} + \cdots + \phi_{J_{n-1}J_n} + \phi_{J_nJ_u}$$
$$= (\phi_{J_u} - \phi_{J_1}) + (\phi_{J_1} - \phi_{J_2}) + (\phi_{J_2} - \phi_{J_3}) + \cdots V_{source} + \cdots + (\phi_{J_{n-1}} - \phi_{J_n}) + (\phi_{J_n} - \phi_{J_u})$$
$$\psi_{BC} = V_{source}$$

Voltmeter is able to read an external voltage source correctly

All contact potentials finally cancel out and are not required to be accounted in circuit equations (provided all are at the same temperature)

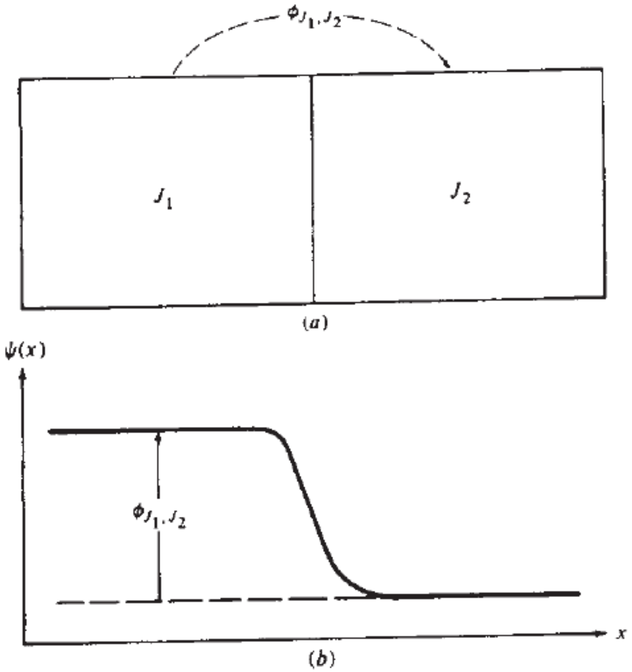
Contact potentials are invisible !!!

Physical Significance of ϕ_{ms} (2)

- In SSD we cannot ignore ***contact potential***

- When one material is p-type and other material is n-type, the built-in potential is a manifestation of contact potential
- When one material is metal and other semiconductor, the Schottky potential barrier is a manifestation of contact potential

- Note, we cannot measure these contact potentials (potential barrier) using voltmeter as seen in the previous problem
- We use indirect approaches such as capacitance to estimate contact potential (potential barrier)



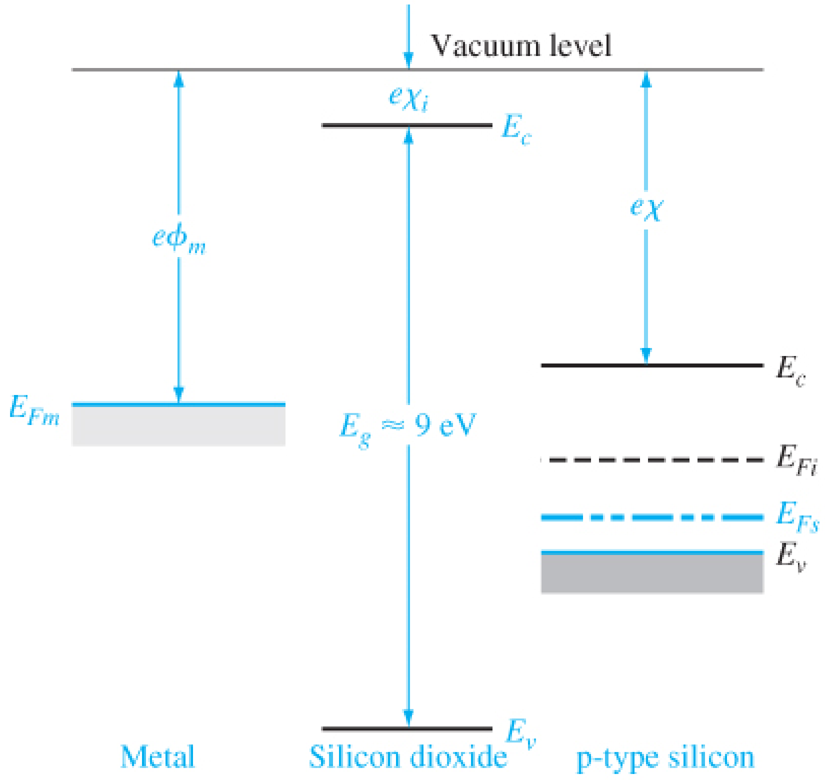
- Concept of contact potential is general
- We estimate contact potential in a MOSFET

MOSFET

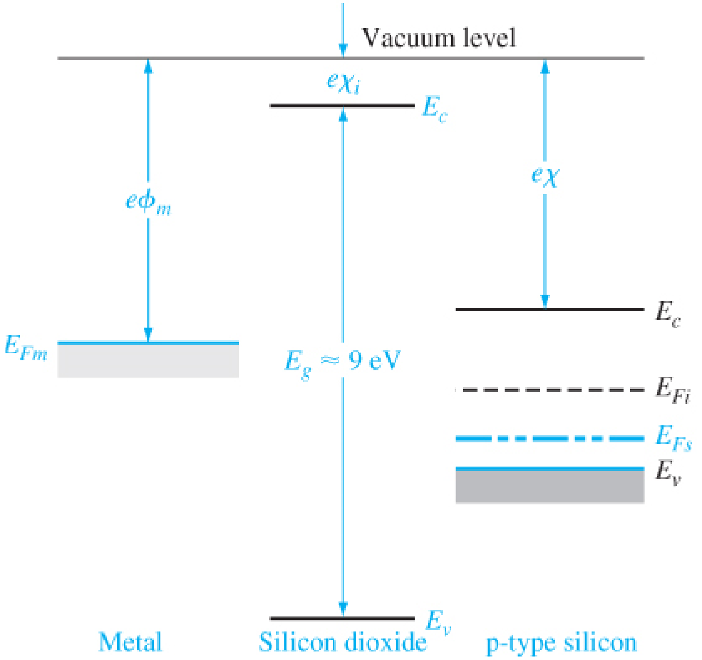
Contact Potential

MOS Capacitor: Material Properties

- Isolated metal, oxide, semiconductor:
  - Metal: work function ϕ_m
  - Oxide: bandgap and electron affinity χ_i
  - Semiconductor: bandgap, electron affinity χ and work function ϕ_s

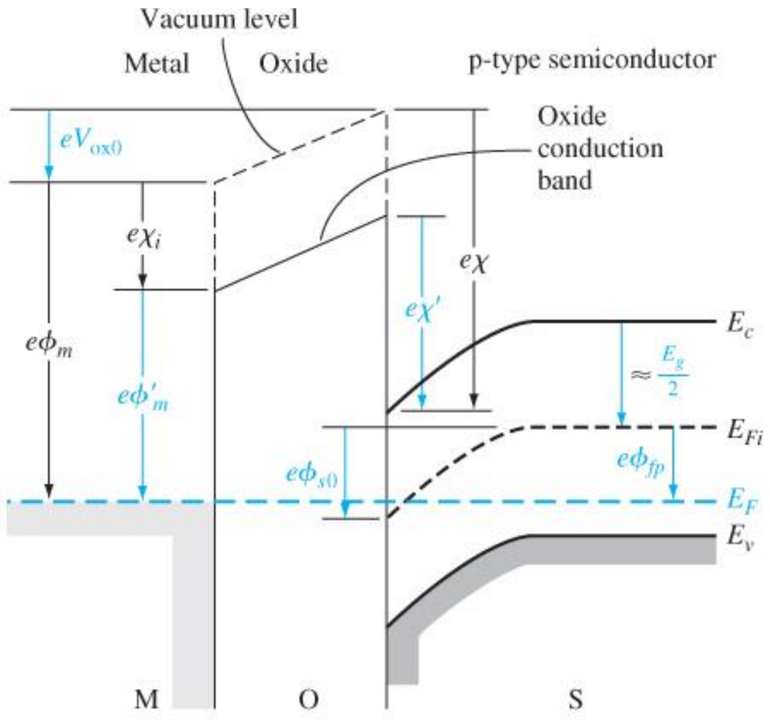


MOS Capacitor: Drawing Band Diagram



1. Fermi level line up in equilibrium (it is flat)

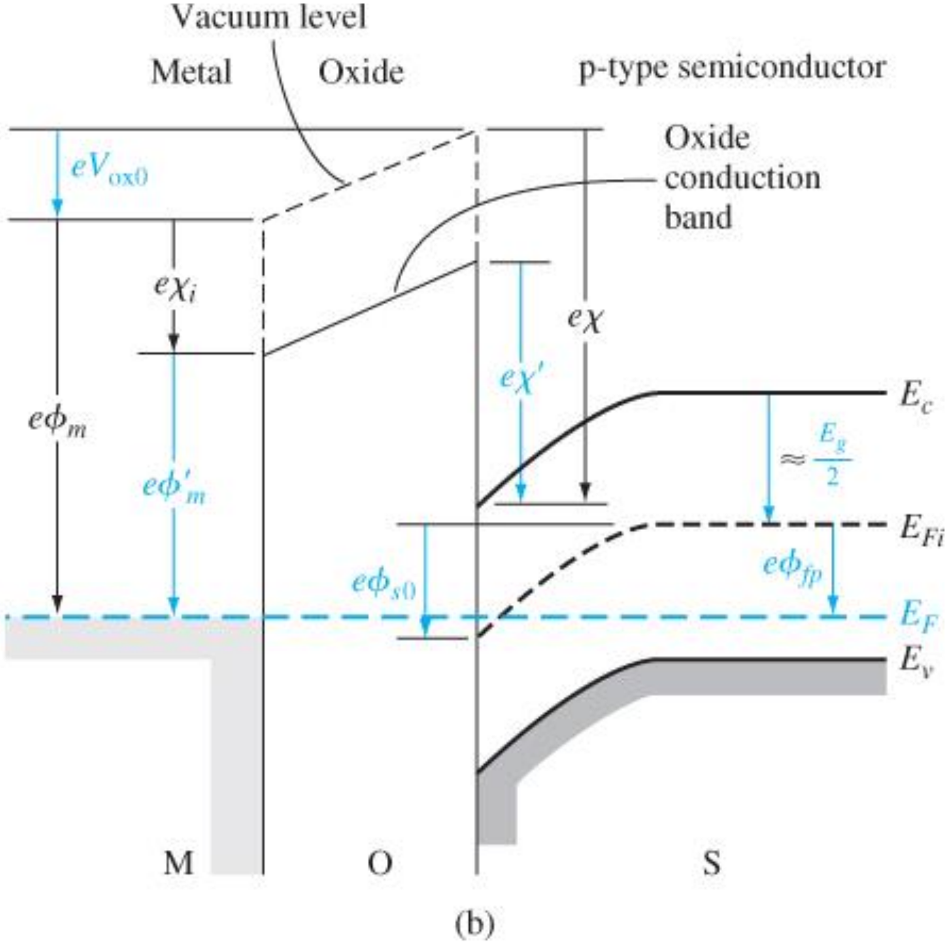
2. When a material changes, the vacuum level is continuous at the interface
   - Because at the interface, the electron is free from one material and also free from other material



(b)

3. Electron affinity is a material property. It does not change with location and doping.
   - As conduction band edge changes, the vacuum level tracks it to keep electron affinity same

MOS Capacitor: ϕ_{ms} (p-type)



We define metal-semiconductor work function difference as:

$$\phi_{ms} = \phi_m - \phi_s = \phi_m - \left(q\chi + \frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$

We can also write (Subtracting $q\chi_i$ on both sides)

$$\phi_{ms} = \phi_m - \phi_s$$
$$= (\phi_m - q\chi_i) - \left(q\chi - q\chi_i + \frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$
$$= \phi'_m - \left(q\chi' + \frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$

where ϕ'_m is potential required to inject electron from metal to the conduction band of oxide and χ' is the modified electron affinity

MOS Capacitor: ϕ_{ms} (n-type)

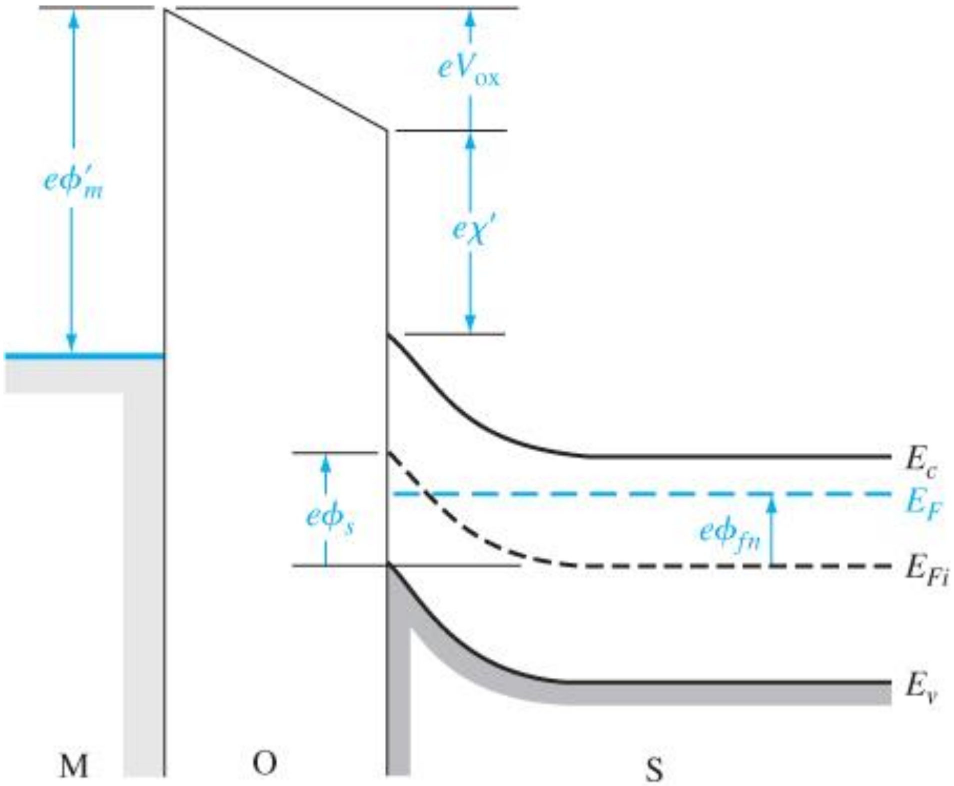


Figure 10.15 | Energy-band diagram through the MOS structure with an n-type substrate for a negative applied gate bias.

We define metal-semiconductor work function difference for n-type semiconductor:

$$\phi_{ms} = \phi_m - \phi_s = \phi_m - \left(q\chi + \frac{E_{g,Si}}{2} - q\phi_{Fn}\right)$$

$$\phi_{ms} = \phi'_m - \left(q\chi' + \frac{E_{g,Si}}{2} - q\phi_{Fn}\right)$$

MOS Capacitor: polysilicon gate

- In mainstream CMOS VLSI technology:
  - n^+ polysilicon gate is used for NMOS (Fermi level of gate close to conduction band)
  - p^+ polysilicon gate is used for PMOS (Fermi level of gate close to valence band)
- Allows source and drain to be self-aligned to gate (eliminates parasitics from overlay errors)
- Obtains threshold voltage of smaller magnitude

In NMOS (body p-type) with n^+ polysilicon gate:

$$\phi_{ms} = \phi_m - \left(q\chi + \frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$
$$= q\chi - \left(q\chi + \frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$
$$= -\left(\frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$

In PMOS (body n-type) with p^+ polysilicon gate:

$$\phi_{ms} = \phi_m - \left(q\chi + \frac{E_{g,Si}}{2} - q\phi_{Fn}\right)$$
$$= q\chi + E_{g,Si} - \left(q\chi + \frac{E_{g,Si}}{2} - q\phi_{Fn}\right)$$
$$= \frac{E_{g,Si}}{2} + q\phi_{Fn}$$

MOS Capacitor: ϕ_{ms}

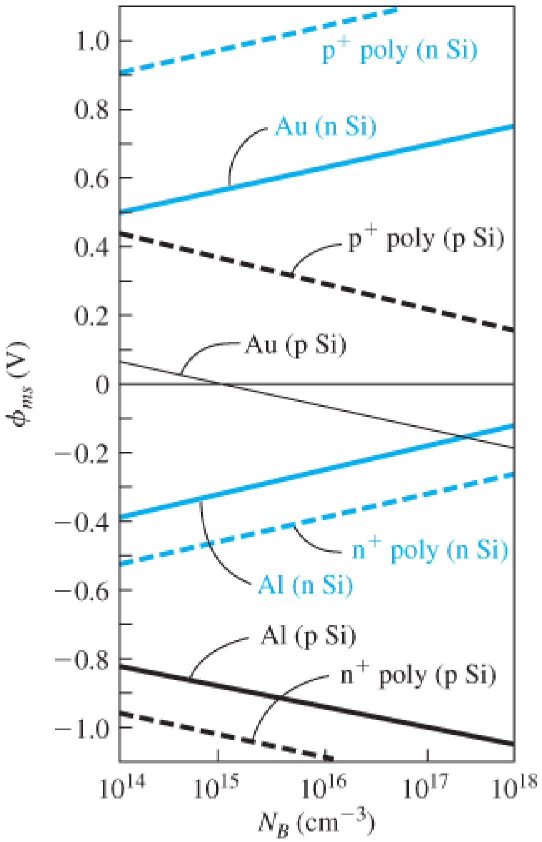


Figure 10.16 | Metal–semiconductor work function difference versus doping for aluminum, gold, and n⁻ and p⁻ polysilicon gates.
(From Sze [17] and Werner [20].)

Work function difference for p-type semiconductor:

$$\phi_{ms} = \phi_m - \phi_s = \phi_m - \left(q\chi + \frac{E_{g,Si}}{2} + q\phi_{Fp}\right)$$

Work function difference for n-type semiconductor:

$$\phi_{ms} = \phi_m - \phi_s = \phi_m - \left(q\chi + \frac{E_{g,Si}}{2} - q\phi_{Fn}\right)$$

References

- Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.
- D. A. Neaman, *"Semiconductor Physics and Devices"*